Battery Failure Boundaries

A Comparative EDA of Li-ion NMC, LFP, and Solid-State Cell Chemistries

Team Members: Jarrett, Aaron | Le, Michael | Medina, Phillip | Zaidi, Hussain | UC Berkeley MIDS

Dataset: battery_failure_surfaces.csv | 60,600 observations × 9 variables

Li-ion NMC

LFP

Solid-State

Research Questions

Q1 Capacity Degradation

Which cell chemistry experiences the lowest capacity fade rate cycle-over-cycle, and when does each chemistry cross the 80% End-of-Life (EOL) threshold?

Q2 Resistance Growth

Which chemistry shows the most favorable (slowest) internal resistance growth over its operational lifespan?

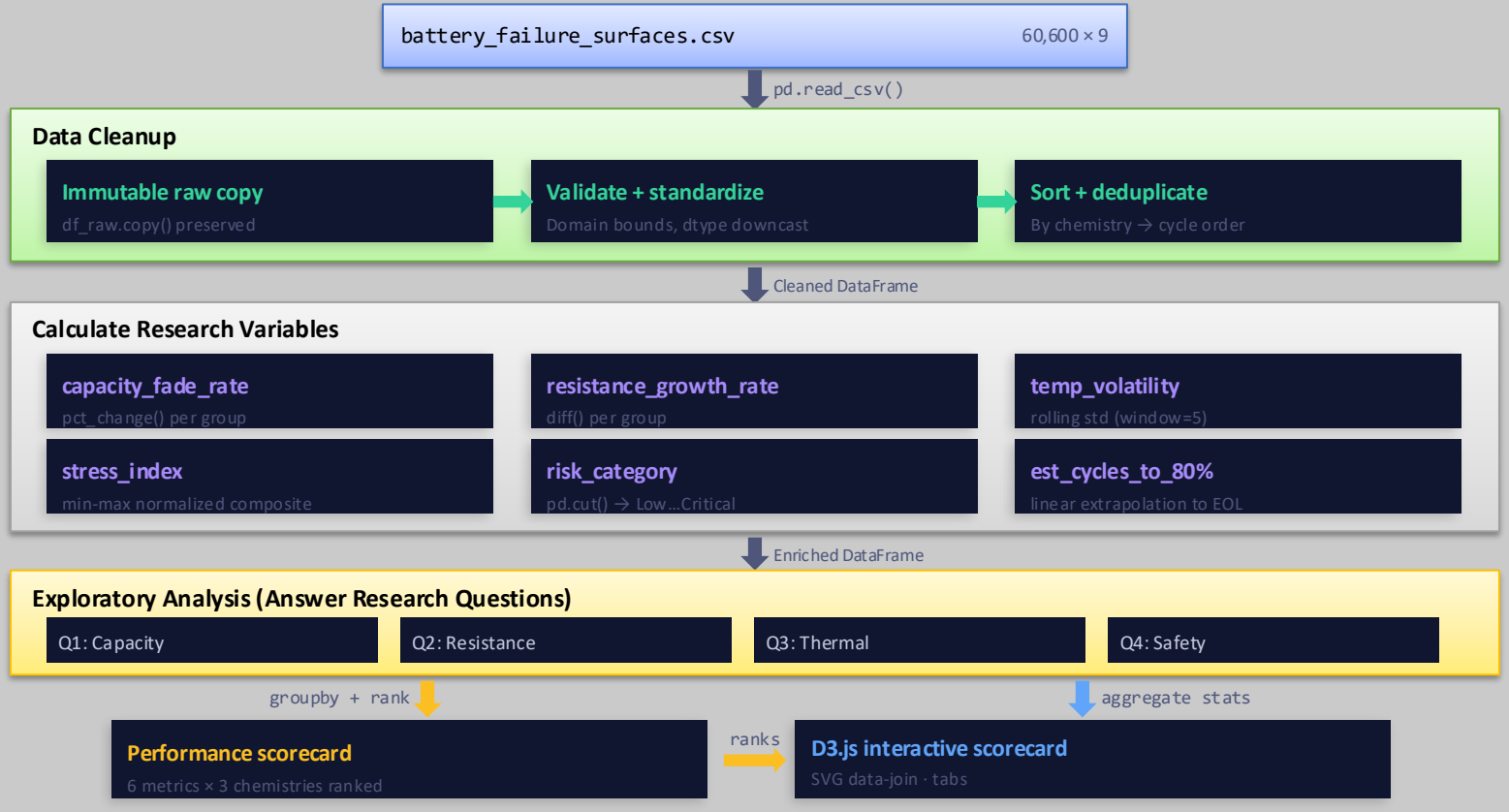
Q3 Safety & Reliability

How do the three chemistries compare across thermal runaway risk, irreversible damage index, and B10 fleet reliability milestones?

Q4 Thermal Stress Sensitivity

Does operational temperature stress (Cool / Optimal / Warm / Hot zones) differentially degrade retention across chemistries?

Data Pipeline



Dataset & Cleaning Overview

60,600

Total Observations

9

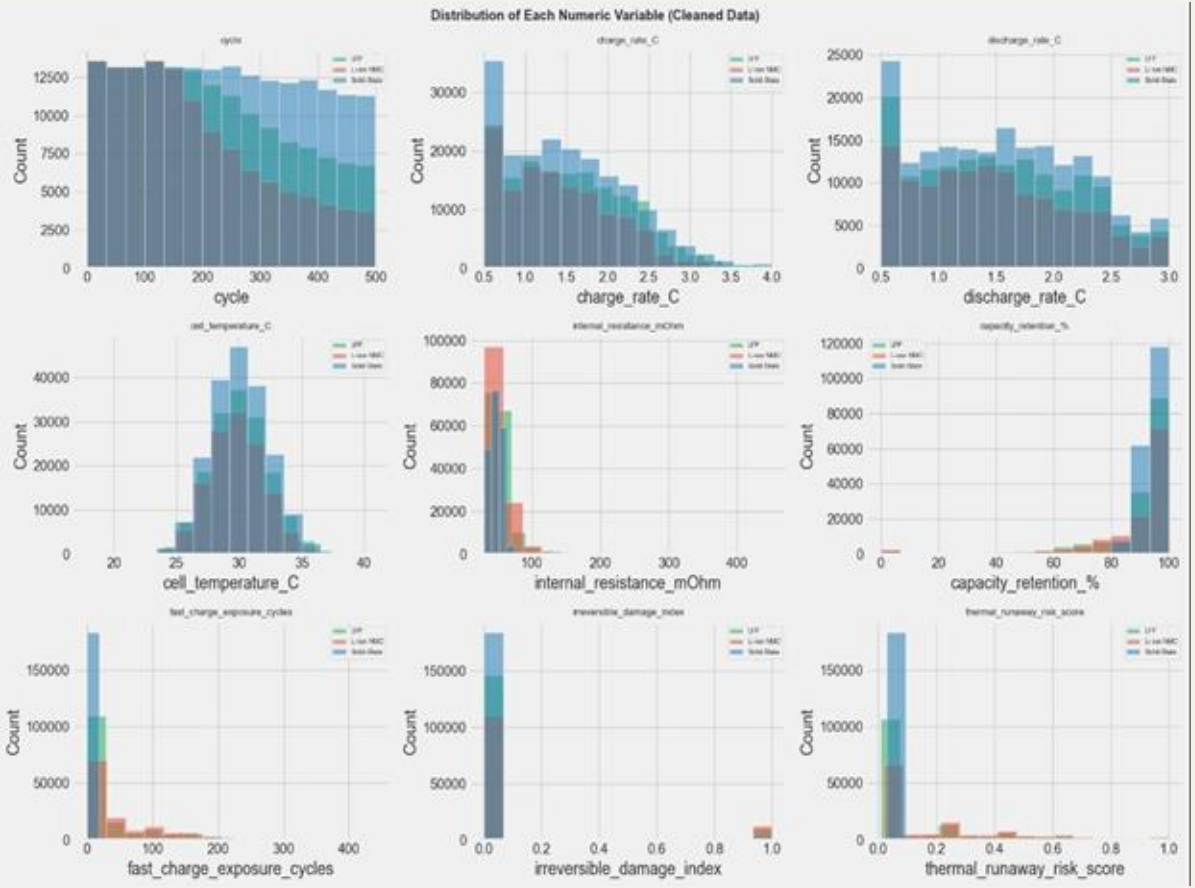
Variables

3

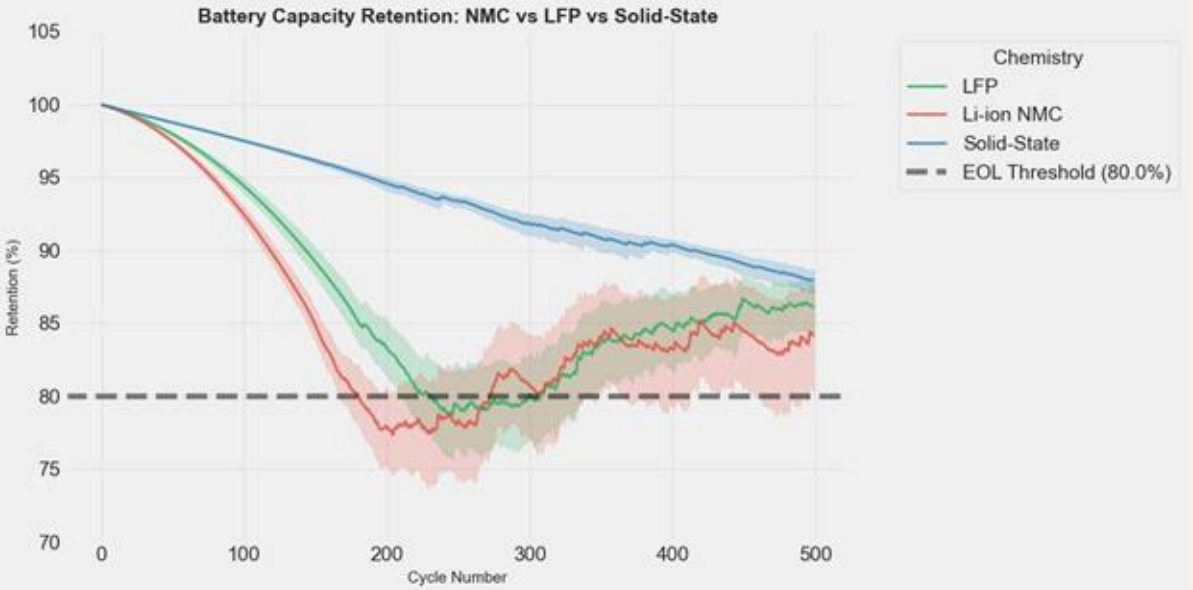
Cell Chemistries

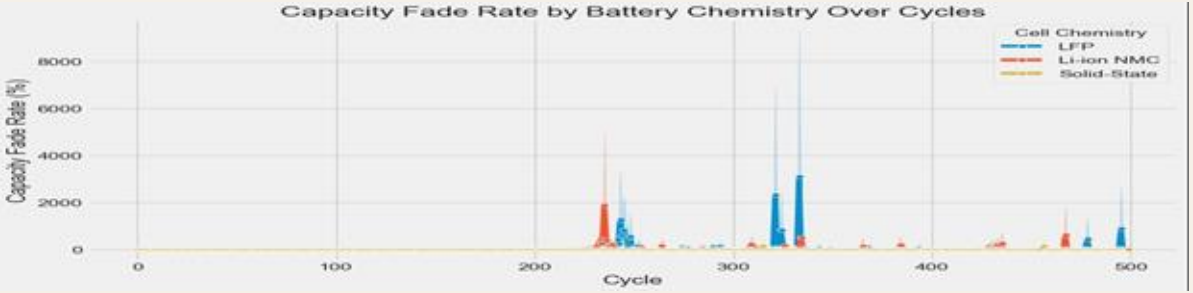
0

Missing Values



Q1: Capacity Degradation Over Cycles





Key Findings

Solid-State

Never crosses the 80% EOL threshold across the full 500-cycle window. Retains ~88% at cycle 500.

LFP

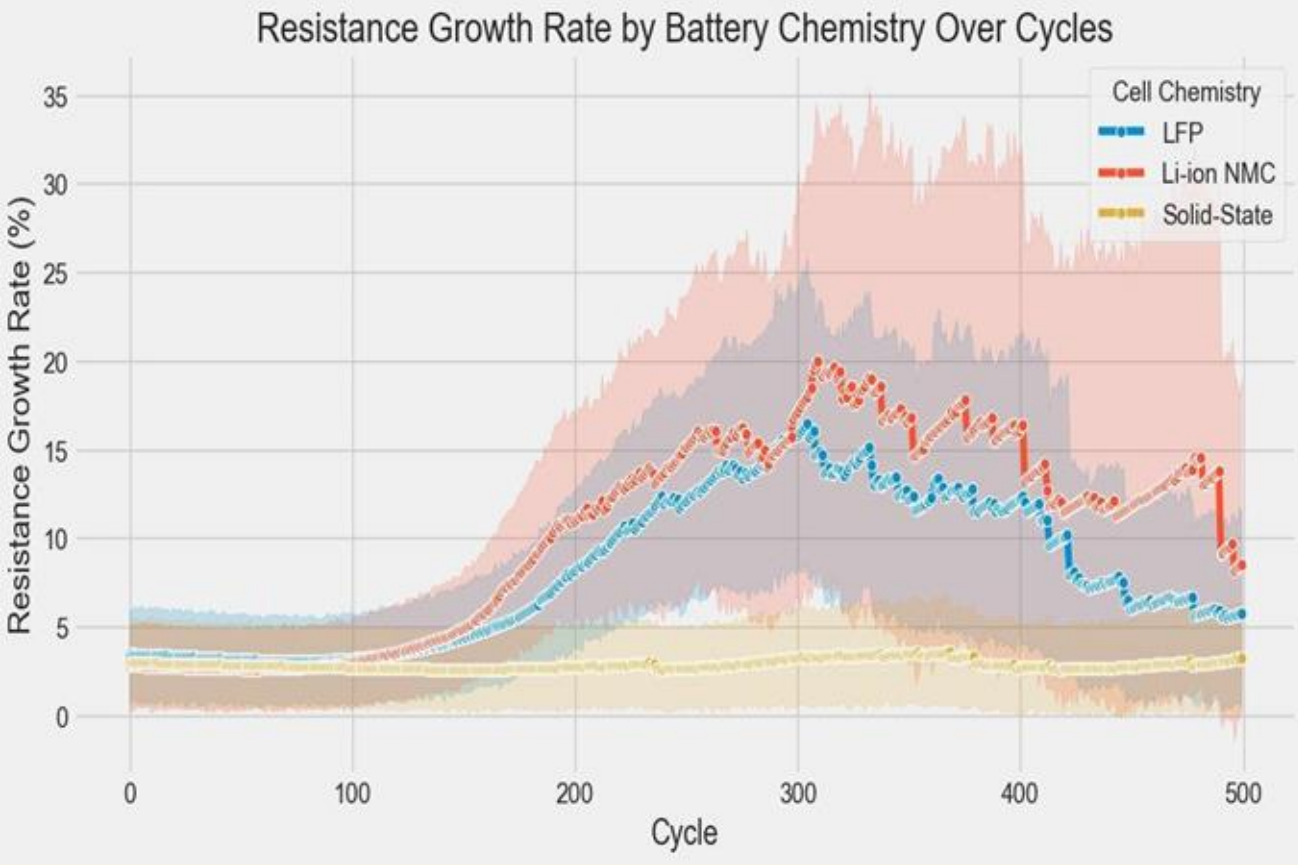
Dips below 80% EOL around cycles 200-300 before recovering. Volatile mid-cycle behavior.

Li-ion NMC

Also dips below 80% EOL in the mid-cycle range. Highest fade rate spikes visible in cycles 200-350.

Solid-State is the only chemistry that never crossed the EOL threshold across the full test window.

Q2: Internal Resistance Growth Over Cycles



Why It Matters

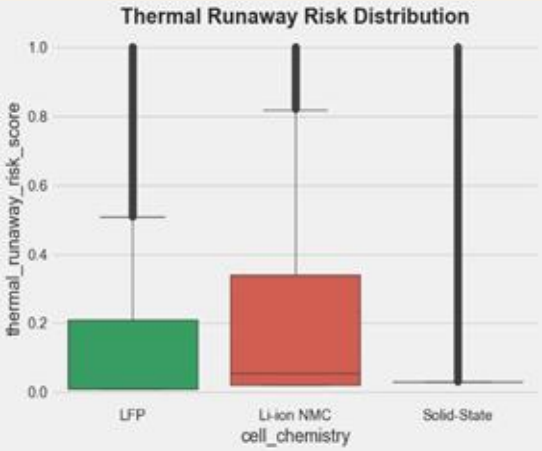
Rising internal resistance means more heat generated per cycle, accelerating degradation and reducing usable power output.

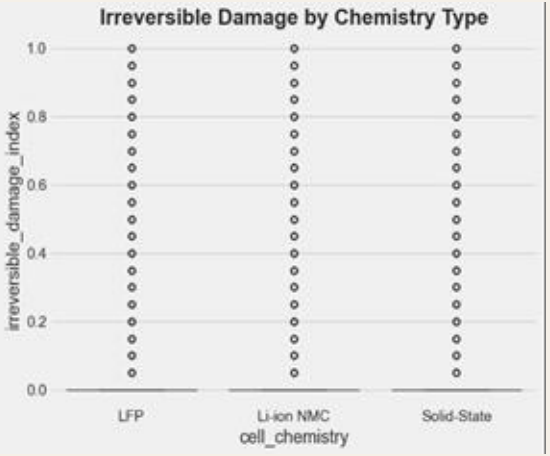
Solid-State
Nearly flat across all 500 cycles. Resistance growth rate stays below 5% throughout.

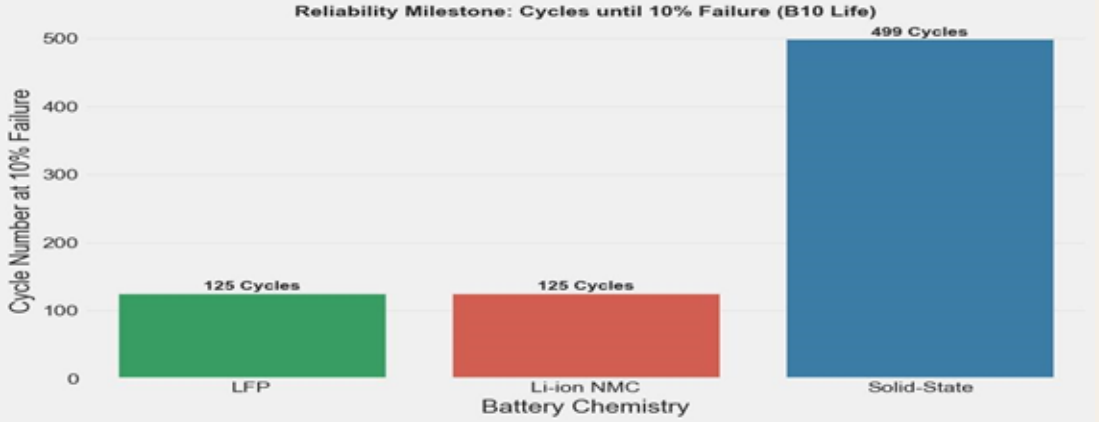
LFP
Moderate growth, rising sharply around cycle 200-300 before declining back.

Li-ion NMC
Largest growth — peaks near 20% around cycle 300 with wide variance. Highest thermal stress risk.

Q3: Safety Profile & Fleet Reliability (B10 Life)







Safety & Reliability

B10 Life

Solid-State:	499 cycles
LFP:	125 cycles
Li-ion NMC:	125 cycles

Thermal Risk Profile

Solid-State

Nearly zero thermal runaway risk across all cycles

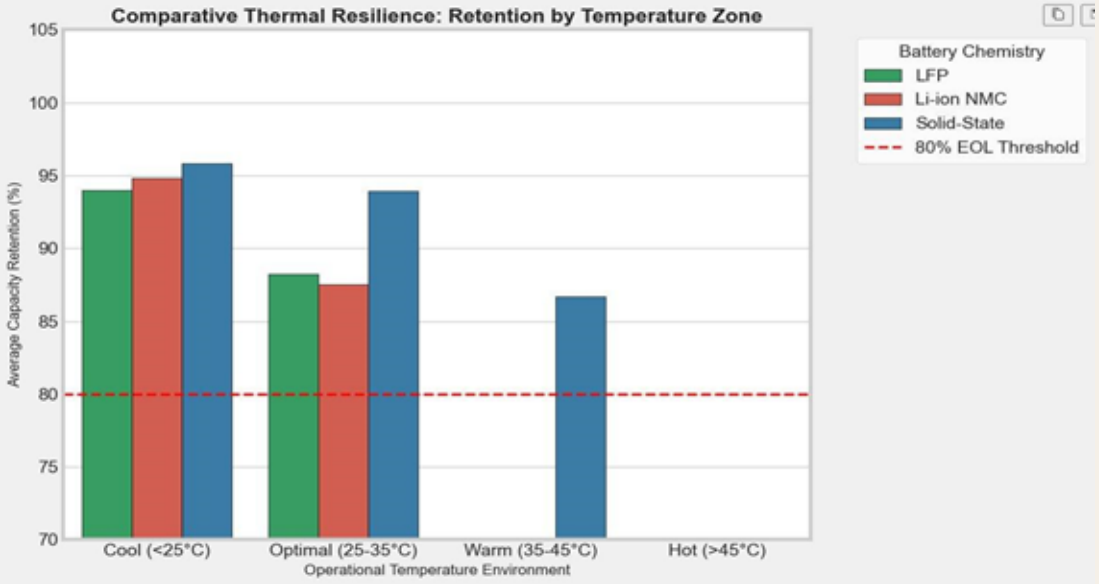
LFP

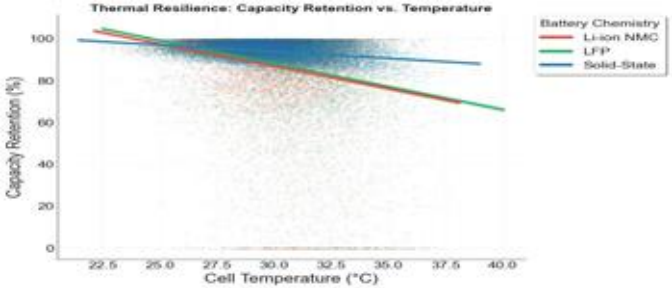
Concentrated IQR at low risk; some extreme outliers

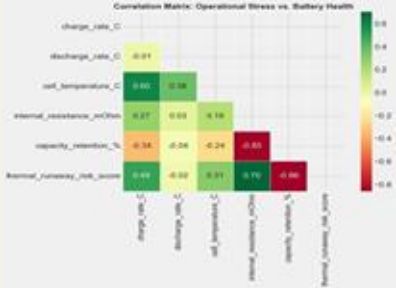
Li-ion NMC

Widest IQR; highest median risk of the three chemistries

Q4: Thermal Stress Sensitivity & Electrochemical Drivers







Thermal Findings

- LFP and NMC have no data in the Warm (35-45°C) or Hot (>45°C) zones, only Solid-State operates at those temperatures in this dataset
- Solid-State retention declines progressively across temperature zones: ~96% cool, ~94% optimal, ~87% warm, still above EOL throughout
- Correlation matrix: internal resistance is the strongest negative driver of capacity retention (r = -0.85). Temperature has a moderate negative effect (r = -0.24)
- Scatter plot confirms Solid-State holds the shallowest slope across temperature, while NMC and LFP decline more steeply

Assumptions & Limitations

Assumption 1

Data treated as representative of laboratory cycling conditions. Real-world duty cycles include variable charge/discharge patterns not captured here.

Assumption 2

pct_change() for fade and resistance rates assumes uniform cycle spacing. Non-uniform intervals would require interpolation prior to rate calculation.

Assumption 3

The dataset is representative of the theoretical performance of each battery chemistry under comparable operating conditions, including cycle life, temperature range, and charge/discharge rates.

Limitation 1

LFP and NMC have no observations above 35°C in the dataset. Thermal zone comparisons for those chemistries are incomplete.

Limitation 2

B10 analysis shows LFP and NMC reach 10% fleet failure by cycle 125, well within the test window. Longer observation past 500 cycles would clarify long-run Solid-State behavior.

Limitation 3

This is descriptive EDA only. No statistical testing was applied. Observed differences between chemistries should not be interpreted as statistically significant.

D3 Visualization

Chemistry Performance *Scorecard*

Conclusions

Across all four research questions, a consistent hierarchy emerged:

Solid-State

Best performer on every metric: never crosses EOL threshold, near-flat resistance growth, lowest thermal risk (near zero), and B10 life of 499 cycles vs. 125 for both competitors.

LFP

Middle performer: dips below EOL in mid-cycles before recovering, moderate resistance growth, and B10 failure at cycle 125. No data above 35°C limits thermal zone conclusions.

Li-ion NMC

Comparable to LFP on most metrics but with higher resistance growth variance and higher thermal runaway risk. B10 failure also at cycle 125. Most volatile fade rate behavior.

This EDA describes patterns observed in the dataset. Results inform chemistry selection decisions, not population-level claims.